

JATIN MALLARAPU

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Summary

Results-driven Full Stack and Java Software Developer skilled in building scalable web applications and microservices using Java, Spring Boot, React.js, Python, and FastAPI. Experienced in REST APIs, SQL databases, OOP, and deploying secure, maintainable solutions on AWS and Azure.

Skills

Programming Languages: Java, JavaScript, TypeScript, Python, SQL, R

Frameworks & Web: Spring Boot, Spring MVC, Spring Security, Spring Cloud, React.js, Angular, HTML5, CSS, Thymeleaf

APIs & Architecture: RESTful APIs, GraphQL, J2EE, JSP, JDBC, Microservices, WebSockets

Messaging & Streaming: Apache Kafka, ActiveMQ, RabbitMQ

Databases: MySQL, PostgreSQL, MongoDB, Amazon RDS, SQL Server

AI Technologies: Spring AI, LangChain, LLMs, RAG Pipelines, NLP, Neural Networks, Prompt Engineering

Tools & DevOps: AWS, Azure, Docker, Git/GitHub, Jenkins, Maven, Redis, Swagger, Postman, JUnit, Mockito, Testcontainers, Jest, Cypress, Linux

Experience

Graduate Assistant

Aug 2024 – Present

TTU Health Sciences Center

Lubbock, Texas, USA

- Improved the university student portal for 15,000+ users by building a responsive React.js frontend, reducing server calls by 40% via pagination and debouncing, and engineered 10+ Spring Boot and PostgreSQL backend features following the full SDLC.
- Designed and implemented 20+ secure RESTful APIs for authentication using Spring Security and JWT, applying core OOP principles and Java data structures to reduce code complexity by 25% and ensure clean, maintainable architecture.
- Containerized the application using Docker and automated 50+ Jenkins CI/CD pipelines for zero-downtime AWS deployments, monitored health via CloudWatch, and managed 100GB+ of structured data using JDBC and Amazon RDS PostgreSQL for reliable storage.

AI Intern

Jun 2022 – July 2024

AIR CENTER VIT AP

Amaravati, Andhra Pradesh, India

- Developed a CNN-BiLSTM model with an Attention mechanism for toxic language detection, implementing a full NLP pipeline in Python with text preprocessing, tokenization, and GloVe embeddings.
- Engineered and optimized model architecture to achieve 99.41% accuracy, performing hyperparameter tuning and embedding refinements to enhance detection of nuanced toxic language in diverse datasets.
- Validated model performance against 5,000+ real-time user comments using JUnit test cases, identifying patterns, correcting errors, and iteratively reducing false positives by 15% for production readiness.
- Delivered a fully production-ready model integrated into the VIT-AP chatbot, enabling accurate real-time moderation and improving user experience through advanced toxic language detection capabilities.

Projects

Scalable Multi-Vendor Microservices Delivery Platform

- Architected and developed a scalable multi-vendor delivery platform using Spring Boot and Spring Cloud microservices architecture to independently manage restaurant, order, and delivery services, ensuring modularity and high scalability.
- Designed and implemented a responsive frontend using React.js and TypeScript, integrating WebSockets to enable real-time order tracking and seamless user experience across dynamic updates.
- Secured the platform using Spring Security with JWT authentication, role-based access control, and refresh tokens, ensuring robust authentication and authorization mechanisms across services.
- Enhanced system reliability and deployment by integrating Kafka for real-time notifications, implementing automated backend testing with JUnit and Mockito, containerizing services using Docker, and deploying via CI/CD pipelines on AWS and Azure.

AI-Powered Smart Healthcare Management System

- Built a healthcare platform using Spring Boot microservices integrated with Eureka and an API Gateway to streamline request routing and improve overall system efficiency.
- Incorporated a Spring AI RAG-based chatbot to deliver medical insights, along with a FastAPI microservice leveraging fine-tuned deep learning models for cancer detection.
- Handled structured patient data using MySQL and JDBC while maintaining clean version control practices through Git-based workflows.
- Deployed the platform on Azure by provisioning and configuring cloud resources to ensure high availability, stable system performance, efficient load handling, and reliable inter-service communication across distributed microservices.

Full-Stack Community Management System

- Designed a unified resident platform using Spring Boot and Spring Security to centralize management of residents, events, and payment workflows in a streamlined system.
- Created interactive dashboards using JSP, Thymeleaf, and CSS, integrating WebSockets for real-time chat functionality while enforcing secure access across application routes.
- Implemented role-based access control and CSRF protection to ensure secure user interactions and maintain data integrity throughout the platform.
- Containerized the application using Docker for consistent deployment environments and coordinated version control and team workflows through Git.

Education

Master of Science in Computer Science

Aug 2024 - May 2026

Texas Tech University, Lubbock, Texas

GPA: 3.8/4.0

B.Tech in Computer Science and Engineering with Specialization in AI

Sept 2020 - Jun 2024

Vellore Institute of Technology, Amaravati, Andhra Pradesh

GPA: 8.75/10.0

Certifications

- **Master Microservices with Spring Boot and Spring Cloud (udemy):** Mastered designing and orchestrating scalable microservices using Spring Boot, Spring Cloud, Docker, and Kubernetes, implementing API Gateways, dynamic service discovery, Fault Tolerance, and distributed tracing.
- **Devops Tools and AWS for Java Microservice Developers (udemy):** Gained expertise in deploying Java microservices using Docker, Kubernetes, Jenkins, and Git, alongside mastering AWS services including EC2, Elastic Beanstalk, Auto Scaling, and Elastic Load Balancer.

Achievements And Awards

- **UWC Paul Whitfield Horn Graduate Fellowship (Link):** Awarded in recognition of academic excellence and potential for professional growth, providing financial support and tuition waiver for the 2025-2026 academic year to encourage full-time graduate study and engagement in scholarly and professional development activities at Texas Tech University.
- **Department of Computer Science Scholarship (Link):** Awarded by the Department of Computer Science at Texas Tech University in recognition of academic achievements, providing \$1,000 in financial support, split evenly over the fall and spring semesters of the 2025-2026 academic year to encourage continued full-time graduate study and engagement in scholarly activities at Texas Tech University.
- **Engineering Clinics Certificate of Merit (Link):** Awarded the "Best Project Award" for securing a position in the top 10 among 500 participating teams at VIT-AP University. Recognized for the successful demonstration of the project "Driver Drowsiness Detection" during the Winter 2022–23 semester, showcasing technical excellence in engineering innovation and collaborative project development.

Publications

- **Detecting Plastic Covers with Computer Vision and Raspberry Pi for the Protection of Cows (Link):** Accepted for oral presentation at the 2nd International Conference on Algorithms and Computational Theory for Engineering Applications (ICACTEA 2025), organized by Aditya University. The study focuses on developing real-time embedded systems to enhance environmental safety and livestock protection.